Mathematical Foundations for Machine Learning

Quiz Answer Key with Explanations | BITS Pilani WILP M.Tech AIML

Question 1

Consider S = {(1,0), (2,0)} in R^2. Which statement is correct?

Options: a) S is linearly independent but does not span R^2. b) S spans R^2 but is not a basis. c) S is a basis for R^2. d) S does not span R^2.

Answer: d) S does not span R^2.

Why: Both vectors lie on the x-axis, so they are linearly dependent [(2,0) = 2·(1,0)] and span only the x-axis (a 1-D subspace), not all of R^2.

Question 2

Let A be an n×n matrix. Suppose A has n distinct eigenvalues. What can definitely be concluded?

Options: a) A is diagonalizable. b) A is invertible. c) All eigenvectors are orthogonal. d) A is symmetric.

Answer: a) A is diagonalizable.

Why: Eigenvectors corresponding to distinct eigenvalues are always linearly independent. n distinct eigenvalues give n linearly independent eigenvectors, so A = PDP■¹ is possible. (Not necessarily invertible if 0 is one eigenvalue; not necessarily symmetric or orthogonal.)

Question 3

Let U and V be finite-dimensional vector spaces. If $U \subseteq V$ and dim(U) = dim(V), then:

Options: a) U = V b) No conclusion can be drawn c) U has fewer vectors than V d) U is a proper subset of V

Answer: a) U = V

Why: A subspace with the same dimension as the whole (finite-dimensional) space must equal that space — a proper subspace always has strictly smaller dimension.

Question 4

S = {(1,0,0), (0,1,0), (1,1,0)} ⊂ R^3. Which statements are correct? I. S is linearly dependent. II. span(S) is 2-dimensional. III. S is a basis for R^3.

Options: a) I only b) II only c) I and II only d) II and III only

Answer: c) I and II only

Why: (1,1,0) = (1,0,0) + (0,1,0), so the set is linearly dependent (I true). All three vectors have z = 0, so span(S) is the xy-plane, dimension 2 (II true). Since it doesn't span R^3, it cannot be a basis for R^3 (III false).

Question 5

For a symmetric positive definite matrix A, the Cholesky decomposition A = LL■ can be viewed as a special case of:

Options: a) LU decomposition b) None of the above c) QR decomposition d) Eigenvalue decomposition

Answer: a) LU decomposition

Why: Cholesky writes A = LL■ = L·U where U = L■ is upper triangular — it's LU decomposition with the extra symmetry constraint U = L■, which is only possible when A is symmetric positive definite.

Question 6

Consider S = {(1,0), (0,1), (0,0)} in R². Which statement is correct?

Options: a) Linearly independent but doesn't span b) Doesn't span R² c) Basis for R² d) Spans R² but not a basis (dependent)

Answer: d) Spans R² but is not a basis because it is linearly dependent.

Why: (1,0) and (0,1) already span R²; adding (0,0) doesn't extend the span but does make the set dependent (3 vectors in a 2-D space, and the zero vector is always a dependency).

Question 7

■x,y■ = 2x■y■ + 3x■y■ + x■y■ – x■y■ – x■y■. For x = (1,1,1), find ||x||².

Options: A) 5 B) 3 C) 6 D) 4

Answer: D) 4

Why: ||x||² = ■x,x■ = 2(1)(1) + 3(1)(1) + (1)(1) – (1)(1) – (1)(1) = 2 + 3 + 1 – 1 – 1 = 4.

Question 8

Which of the following matrices is in reduced row echelon form (RREF)?

Options: A) [1 0 0 / 0 0 1 / 0 0 0] B) [1 2 3 / 0 1 4 / 0 0 1] C) [1 0 0 / 0 1 0 / 0 1 0] D) [0 1 2 / 1 0 0 / 0 0 1]

Answer: A

Why: RREF requires: leading 1's move strictly right going down, each leading 1 is the only nonzero entry in its column, and zero rows are at the bottom. Only A satisfies all of these (leading 1's in col 1, then col 3, with zero row last, and no other nonzero entries in those columns). B has nonzero entries above its leading 1's; C repeats a leading column; D's leading entries aren't in increasing column order.

Question 9

After Gaussian elimination: [1 3 -2 | 1], [0 1 4 | -2], [0 0 0 | 0]. Number of variables = 3. What is true about the solution?

Options: a) Infinitely many solutions b) No solution c) Unique solution d) Inconsistent system

Answer: a) Infinitely many solutions

Why: There are 3 unknowns but only 2 independent (nonzero) equations — the last row (0=0) gives no new information. One variable is free, so the system is consistent with infinitely many solutions.

Question 10

Which of the following pairs contain positive definite matrices? (4 answer choices, each with 2 matrices — both matrices in the chosen pair must be PD)

Options: a) [[1,2,0],[2,1,0],[0,0,1]] & [[0,1,0],[1,0,1],[0,1,0]] b) diag(2,3,4) & [[3,0,0],[0,3,1],[0,1,3]] c) [[4,1,0],[1,4,0],[0,0,4]] & [[1,0,0],[0,1,0],[0,0,-2]] d) [[2,1,0],[1,2,0],[0,0,2]] & [[3,1,1],[1,3,1],[1,1,3]]

Answer: d) (Note: option b is also mathematically valid — see note below)

Why: Checking eigenvalues/leading principal minors: (a) first matrix has a negative leading 2×2 minor (1·1–2·2 = –3), not PD. (c) second matrix has a –2 eigenvalue, not PD. (d) both matrices have all-positive leading principal minors (and eigenvalues {1,2,3} and {2,2,5}) — both PD.

■ *Heads up: I double-checked this one with actual eigenvalue computation. Both pair (b) and pair (d) turn out to be genuinely positive definite (all eigenvalues positive in both cases). This looks like a flawed question with two technically correct options. I selected (d) as the primary answer, but if it's marked wrong, (b) is the alternative — worth raising with your instructor.*

Question 11

What is the definition of the trace of an n×n square matrix A?

Options: a) Product of diagonal elements b) Sum of all elements c) Sum of diagonal elements a■■ d) Determinant of the matrix

Answer: c) The sum of the diagonal elements a■■.

Why: By definition, trace(A) = ∑ a■■ (sum over i = 1 to n of the diagonal entries).

Question 12

Which matrix is positive definite? a) [[0,1],[1,0]] b) [[1,0],[0,1]] c) [[1,2],[-2,1]] d) [[1,-2],[-2,4]]

Options: See options above.

Answer: b) [[1,0],[0,1]]

Why: (a) has eigenvalues +1, –1 — not PD. (b) is the identity, eigenvalues 1,1 — PD. (c) is not symmetric (complex eigenvalues) — not PD in the real sense. (d) has determinant 0 (eigenvalue 0) — only positive semi-definite, not strictly PD.

Question 13

The homogeneous system Ax = 0 has nontrivial solutions if:

Options: a) det(A) ≠ 0 b) A is invertible c) rank(A) = n d) det(A) = 0

Answer: d) det(A) = 0

Why: If det(A) ≠ 0, A is invertible and Ax = 0 has only the trivial solution x = 0. Nontrivial solutions exist exactly when A is singular, i.e., det(A) = 0.

Question 14

■u,v■ = 3, ||u|| = 4, ||v|| = 5. Find ||u – v||.

Options: A) 6 B) √35 C) 2 D) √28

Answer: B) √35

Why: ||u–v||² = ||u||² – 2■u,v■ + ||v||² = 16 – 6 + 25 = 35, so ||u–v|| = √35.

Question 15

B = {(1/√2)(1,1), (1/√2)(1,-1)} is an orthonormal basis for R². Find the coordinates of v = (3,4) in basis B.

Options: a) (7/√2, –1/√2) b) (7/√2, 1/√2) c) (5/√2, 1/√2) d) (–1/√2, 7/√2)

Answer: a) (7/√2, –1/√2)

Why: For an orthonormal basis, each coordinate is v·b■. c■ = v·b■ = (3+4)/√2 = 7/√2. c■ = v·b■ = (3–4)/√2 = –1/√2.

Question 16

Let a and b be nonzero vectors with angle 60° between them. What is the angle between a and –b?

Options: a) 120° b) 90° c) 120° only if |a| = |b| d) 60°

Answer: a) 120°

Why: cos(angle between a, –b) = (a·(–b))/(||a||·||b||) = –cos(60°) = –0.5, so the angle is 120°. This holds regardless of the vectors' magnitudes, since lengths cancel out in the cosine formula.

Question 17

Let A be a 3×5 matrix. The homogeneous system Ax = 0 always has:

Options: a) Exactly three solutions b) Only the trivial solution c) No solution d) Infinitely many solutions

Answer: d) Infinitely many solutions

Why: With 5 unknowns and at most rank 3 (only 3 rows), there are at least 5 – 3 = 2 free variables. A homogeneous system is always consistent (x = 0 works), and free variables guarantee infinitely many solutions.

Question 18

By the leading principal minors test, a symmetric matrix A is positive definite if and only if:

Options: Options: all entries positive / det(A)=0 / all leading principal minors strictly positive / all leading principal minors non-negative

Answer: All leading principal minors are strictly positive.

Why: This is Sylvester's criterion: A symmetric matrix is positive definite iff every leading principal minor (determinant of the top-left k×k submatrix, for k = 1,...,n) is strictly greater than zero.

Question 19

Let A ∈ R■×■. The matrix A is diagonalizable over R if and only if:

Options: a) det(A) ≠ 0 b) A has n linearly independent eigenvectors over R c) A has n distinct eigenvalues d) A is symmetric

Answer: b) A has n linearly independent eigenvectors over R.

Why: This is the precise definition/criterion for diagonalizability. Distinct eigenvalues (c) is sufficient but not necessary; symmetric (d) guarantees diagonalizability but isn't required in general; det ≠ 0 (a) only rules out a zero eigenvalue and is unrelated to diagonalizability.

Question 20

If A is symmetric positive definite, then its eigenvalues are:

Options: a) All negative b) All zero c) All positive d) Sum is zero

Answer: c) All positive

Why: By definition, A is positive definite iff x■Ax > 0 for all nonzero x, which for a symmetric matrix is equivalent to every eigenvalue being strictly positive.